

C¹

A course compiler.

One chapter PDF in. A cited concept graph, a lesson, an assessment and four typeset documents out — with every factual sentence traceable to the exact span of the page it came from.

SOURCE	TESTS	PASSES	TARGETS
6,846 lines · 42 modules	247 · 100% core branches	10, each owning its column	4 PDFs + a certificate

In one paragraph

Ask a language model to “make a lesson from this chapter” and you get prose that reads fine and cannot be checked. You cannot point at a sentence and say why it is true, you cannot separate a hallucinated formula from a real one, and re-running it changes the output for reasons nobody can name. **CourseC treats course generation as a compilation problem instead.** The source chapter is the input language; a typed Concept Graph in SQLite is the intermediate representation; ten independently-tested passes transform it; four typeset documents are the code generation step. Every claim that reaches a page carries the citation it was derived from, every number is executed rather than written, and any pass that violates one of seven invariants fails the build rather than degrading quietly.

WHO IT IS FOR

An instructor or curriculum team who has the source material and needs defensible teaching artifacts from it — where “defensible” means a colleague can challenge any sentence and be shown the page it came from, and a wrong number is caught by the build rather than by a student.

The problem, stated precisely

A single prompt produces a **plausible artifact with no audit surface**. Three specific failures follow, and each one is a silent failure — the output looks identical whether or not it happened.

UNFALSIFIABLE PROSE

A claim with no citation cannot be checked, only believed. Reviewing it means re-reading the source yourself — the work the tool was supposed to do.

INVENTED NUMBERS

A model that writes “ $m \cdot a = 5$ ” for $m=2$, $a=3$ is producing text that pattern-matches arithmetic without performing any.

UNSTABLE OUTPUT

Re-running changes the result for untraceable reasons, so nothing can be regression-tested and no fix can be shown to hold.

The thesis

THE MAPPING

source chapter	input language
Concept Graph	the IR
ten passes	compiler passes
four PDFs	code generation
certificate	build log

Compiling instead of prompting buys three things a prompt chain cannot.

A typed intermediate representation. Every fact the system knows is a row in a real schema — a Concept, a SourceSpan, a WebEvidence chunk — not a paragraph one model wrote for another model to re-read. The graph can be queried, diffed, and rendered without asking a model anything.

Passes that own their columns. ingest only ever writes Block and SourceSpan; compose only ever writes LessonBlock. This is enforced at runtime, not documented: `Graph.add()` looks up the declared owner for a row's type and raises `OwnershipViolation` if the writing pass is not it. A pass genuinely cannot write outside its lane.

Diagnostics instead of vibes. Every pass reports through a structured `Diagnostic` carrying severity, code and message, rather than printing or swallowing problems. One error-severity diagnostic anywhere means the build emits no PDF at all.

The seven invariants

These are not aspirations in a design document. Each one is the thing an adversarial test exists to break, and each has a named mechanism behind it.

INVARIANT	WHAT ENFORCES IT
I1 No unsupported sentence. Every emitted sentence carries a citation and passes entailment against the spans it cites.	A critic that sees one sentence and only that sentence's own cited spans , looked up fresh by id — never the dossier that produced it. Fails → repair twice → drop → quarantine the slot.
I2 Numbers are computed, never generated. Every numeric literal traces to a span, a cited source, or a sandbox result.	A worked example's formula , substitutions and claimed_result are executed as SymPy in a subprocess with a timeout, and compared. A linter then scans surviving prose for stray numerals.
I3 Provenance is total. Every node reachable from a document traces to file or URL, locator, retrieval time and content hash.	SourceSpan carries the file, page, bbox , char_range and sha256 , with a running character offset across the whole document — byte ranges, not "somewhere in the PDF".
I4 The prerequisite graph is acyclic. Cycles break by a deterministic, logged policy — never silently.	Repeatedly remove a cycle's lowest-confidence edge, tie-broken on a content hash rather than a random id — a random tie-break would make cycle-breaking irreproducible, defeating the point.
I5 Contract before emission. A concept enters a document only with a contract status attached.	Five slots: definition, intuition, worked example, visual-or-analogy, ≥2 assessment items. A concept short of them is marked partial and lists its unmet slots, never silently emitted as complete.
I6 Contradiction is surfaced, never resolved. The source stays primary.	A numeric mismatch between retrieved evidence and the source writes a contradicts edge and a marked note carrying both citations. The source row is never overwritten.
I7 Licence-clean media only.	An image is a bounding-box crop of the source page, generated diagram code (Mermaid), or nothing. No third image path exists in the code.

THE RULE THAT MAKES THE REST CREDIBLE

A gate that never rejects anything is broken, not perfect. The pipeline asserts its aggregate item-rejection rate is nonzero, and the adversarial suite is strengthened whenever it passes on first write.

The intermediate representation

Everything the compiler knows lives in one SQLite database as typed `SQLModel` tables. Every node carries `id`, `created_by_pass`, `content_hash` and `created_at`; `created_by_pass` is immutable after first write, so provenance of **who created a row** cannot drift after the fact.

NODE TYPES

Block	a classified unit of a page
SourceSpan	exactly where it came from
Concept	definition, formula, procedure, ...
Alias	a name that merged into a concept
SyllabusNode	a curriculum entry to anchor to
WebEvidence	a retrieved, scored, tiered chunk
LessonBlock	one contract slot, with citations
Verdict	the critic's judgement on a sentence
ExecResult	what the sandbox actually computed
Item	an assessment item and its gate history
Misconception	faulty reasoning a distractor encodes
ItemStats	difficulty and discrimination
Mastery	a student's posterior, append-only

OWNERSHIP, ENFORCED

```
>>> graph.add(Block(
    created_by_pass="understand", ...))
OwnershipViolation: Block is owned
by pass 'ingest', not 'understand'
```

The same check covers edges: `Edge.kind` is a closed enum, and each kind declares its owning pass. `prerequisite_of` and `part_of` belong to `structure`; `contradicts` to `evidence`; `assesses` to `assess`. `evidenced_by` is deliberately shared between `evidence` and `compose` — it is the same relationship at two stages, so it gets one kind rather than two near-duplicates.

An edge whose endpoints are not already in the graph is refused outright, which makes a dangling citation structurally impossible rather than merely unlikely.

The pipeline, pass by pass

Run end to end by `coursec build chapter.pdf`. Each pass is independently unit-tested against a scripted model backend — none of the tests need a live model or a network connection.

01 ingest	WRITES Block, SourceSpan PyMuPDF yields raw text and image blocks; a typography heuristic classifies each one — font size against the body-text baseline, bullet prefixes, a regex for <code>Figure N.M</code> captions — into heading, paragraph, figure, caption, table, equation or list. Captions bind to figures by sequence first and geometry second; an unbound figure raises a diagnostic rather than passing silently.
02 understand	WRITES Concept, Alias, SyllabusNode Extraction is one model call per section , never per paragraph and never per pair — a chapter has dozens of paragraphs but a handful of subsections, and calling a model in a loop over more than that is the cost anti-pattern the design forbids. Canonicalisation is then model-free: candidates are embedded locally and agglomeratively clustered. The merge threshold of 0.80 is not a guess — it is tuned against a committed labelled fixture where the pair that must merge (“Ohm’s law” / “ $V = IR$ ”) embeds at 0.855 , and the closest false-positive risks (“variance”/“covariance” at 0.764 , “hypothesis”/“theory” at 0.773) both sit below 0.78 .
03 structure	WRITES prerequisite_of, part_of An edge exists only when two independent signals agree : concept A’s definition text mentions concept B by name, <i>and</i> a pairwise model judgement — run only over candidate pairs, never all $O(n^2)$ — agrees on the direction. Cycles are then broken deterministically (I4).
04 gap	WRITES GapVector, RetrievalBudget (transient) Four fields per concept: coverage, depth, modality and prerequisite. Budget is allocated proportionally under a hard global cap. A fully-covered chapter allocates a budget of zero — deciding not to search is a valid outcome, not a missing feature — and the allocation is provably scale-invariant.

05 evidence	WRITES WebEvidence, evidenced_by, contradicts Real retrieval: <code>httpx, robots.txt</code> honoured via <code>urllib.robotparser</code> rather than parsed and ignored, per-domain rate limiting, and BeautifulSoup stripping navigation before chunking by heading. Admission is policy, not judgement: present-in-source or corroborated by ≥ 2 independent tier-1 domains $\rightarrow$ admitted; a single tier-1 source $\rightarrow$ weak , usable for enrichment but never as the sole citation for a formula or definition ; anything else $\rightarrow$ rejected with a reason.
06 compose	WRITES LessonBlock, evidenced_by Each generative slot is one model call constrained to a <i>dossier</i> : the concept’s source span, its admitted evidence best-first, and one-line prerequisite summaries. If the dossier has no grounding material, generation refuses before the backend is ever called — a code-level guard, not a prompt instruction hoping the model behaves, which is what makes “strip the dossier and confirm it refuses” a deterministic test rather than a hope.
07 verify	WRITES Verdict, ExecResult The critic is blind to everything except the sentence and its own citations (I1). Separately, every worked example is executed (I2). Every judgement — including intermediate repair attempts — is a stored row, not just the final one, so the history of a claim survives.
08 assess	WRITES Item, Misconception, ItemStats One item per concept $\times$ Bloom level, typed by the concept’s kind. Every MCQ distractor must link to a real misconception node , sourced from sentences the critic actually marked contradicted. Four blocking gates run in full even after an early failure, so the rejection log shows every reason: independent key verification (5 samples, majority must agree), leakage (answerable with no course context means it tests trivia), single-answer (every distractor must be defensibly wrong), and numeric execution. Survivors face a synthetic pilot: 12 simulated students on a fixed ability grid, and a from-scratch 2PL IRT fit flags degenerate items. This is labelled <i>screening</i> , never calibration — twelve students cannot support a calibration claim, and the wording is not allowed to drift.
09 emit	WRITES rendered targets Typst markup assembled from the graph. The cheat sheet is a pure template over each concept’s own span, so it makes zero model calls — asserted on the cache-miss counter. The question paper selects greedily against a blueprint of marks, Bloom mix and unit weightage; an infeasible blueprint names its binding constraint rather than returning something merely close. A fixed compile timestamp makes “identical IR $\rightarrow$ byte-identical PDF” a checked property.
10 learn	WRITES Mastery Bayesian Knowledge Tracing as one closed-form update, pure and free of I/O so a property test can hold it to the textbook formula directly. A wrong answer does not stop at “this concept is weak”: <code>diagnose_root_cause</code> walks prerequisite edges upstream, following the weakest still-weak prerequisite, and stops at the deepest one — because remediating a concept is wasted effort if the real gap is two levels below it.

The stack, and why each piece

CHOICE	WHY THIS ONE
SQLModel + SQLite	The IR must be queryable, diffable and inspectable without a server. SQLModel gives Pydantic validation and SQL in one type, so the schema is the documentation. One file is the whole build output.
PyMuPDF	Gives per-block bounding boxes and font metrics, which is what makes both the typography classifier and licence-clean figure cropping (I7) possible. A plain text extractor would lose exactly the data the invariants need.

sentence-transformers	Canonicalisation and retrieval scoring run locally , so the expensive, non-deterministic model is reserved for judgement calls and never used for similarity.
SymPy	Executes a claimed calculation in a subprocess with a timeout. This is what turns I2 from a promise into arithmetic that either agrees or does not.
Typst	Programmatic typesetting with a deterministic compile — the fixed-timestamp property that makes byte-identical output testable. A LaTeX toolchain would be heavier and harder to pin.
Anthropic API	The one production model backend, injected as an explicit argument everywhere. It fails loudly when unconfigured and never fabricates — there is no code path where a pass can quietly reach for a live model.
FastAPI + uvicorn	A read-mostly HTTP surface over the graph. Chosen for typed request models that match the Pydantic already in use, so the API shares the IR’s validation rather than duplicating it.
httpx + BeautifulSoup	Real fetching with real politeness — robots.txt, rate limits, navigation stripping — because “retrieval” that ignores those is a demo, not a system.
Hypothesis	Property tests over generated input, for the invariants that must hold for any graph rather than one fixture.
Typer · ruff · uv	A CLI that documents itself, one linter, and reproducible locked environments.

How it is proven

Four kinds of test, and every pass writes whichever apply.

UNIT

Pure functions and schema validation — domain-tier classification, gap-vector arithmetic, the BKT step.

GOLDEN

A committed snapshot of ingest against a real, openly-licensed fixture chapter. Diffs are reviewed, never blanket-regenerated.

PROPERTY

Invariants over generated input: **the prerequisite graph is acyclic after cycle-breaking, for any graph.**

ADVERSARIAL

Poisoned input the system must catch on purpose: a low-authority page asserting a wrong constant, an MCQ with two defensible answers, arithmetic that does not hold.

Above those sits a **demo harness** (coursec demo) that answers a different question than the per-pass tests do: not “does each pass catch its own defect in isolation”, but “is the same planted defect still caught once the real passes are wired together the way a build wires them”. Five scenarios, each scored against the specific diagnostic code its invariant promises to raise — never against “the build did not crash”. A scenario that raises an unexpected exception is reported as its own *error* status, distinct from a scenario that ran clean and failed to catch its defect, so an environment problem can never be mistaken for a regression.

BY THE NUMBERS

tests	247
core branch coverage	100%
source lines	6,846
test lines	4,359
modules	42

Coverage is enforced at 100% on the core IR specifically, where a silent regression would corrupt every pass downstream.

What ships

COMMAND LINE

build compile a chapter end to end
web serve the frontend and its API
quiz adaptive session in the terminal
serve the Streamlit quiz
demo the adversarial harness

THE WEB SURFACE

Four views over a live graph: an overview, a concept-graph explorer laid out by prerequisite depth, the adaptive quiz, and the invariant ledger. The quiz endpoints call the learn pass rather than reimplementing its mathematics in a request handler, and an item’s key is never sent to the browser until an answer is in.

DOCUMENTS

Booklet	every contract slot, diagrams rendered at build time, figures cropped by stored bbox, bibliography walked from citation edges
Cheat sheet	a pure template over source spans — zero model calls
Question paper	greedy selection against a blueprint
Answer key	reads stored execution results rather than recomputing
Certificate	written either way — it is the artifact that explains a failure

The certificate view is **recomputed, never replayed**: a build’s diagnostics are not persisted, so the API derives each row from the stored verdicts, execution results and gate outcomes — and says so in its own response rather than implying it read a log it never saw.

DESIGN LANGUAGE

Editorial Ink — warm paper, one oxblood accent, and two semantic hues at matched lightness. The organising idea is that **the margin is a first-class column**, because marginalia is precisely what this compiler’s output is: a claim in the body lights its own source in the margin. The mark is a serif *C* with a superscript numeral — a compiled claim carries its citation.

What is deliberately not claimed

Credibility depends on being exact about the edges of the work.

LIMIT	THE HONEST VERSION
Search backend	The retrieval machinery is real — fetching, robots, tiering, admission, contradiction — but no search provider is wired in by default. Without one it stops cleanly rather than fabricating results.
Pilot statistics	12 simulated students is a <i>screen</i> for degenerate items, not calibration, and every output string says so.
BKT parameters	Illustrative defaults, not fit to any real cohort. The update rule is the textbook one; the four constants are not evidence about your students.
Demo-grade components	The block classifier, the token-budget proxy and the numeric-mismatch check are heuristics, and each is marked as such in the source rather than dressed up.
Remediation content	Mastery tracking and root-cause diagnosis ship; generating remediation for a diagnosed gap does not. The edge kind is declared and unused, and the README says so.

Where it goes next

NEAR	STRUCTURAL	THE INTERESTING ONE
Wire a search provider, turning correct retrieval machinery into retrieval that actually crawls. Generate remediation against diagnosed root causes, closing the loop the learn pass opens.	Multi-chapter builds, where the prerequisite graph spans a whole course and the distance from a concept to its foundations becomes a curriculum-design signal rather than a per-chapter one.	Real cohort data makes the pilot a calibration rather than a screen, and the BKT constants something fitted rather than assumed — at which point the honest caveats above can be retired one at a time, with evidence.

C¹ *This document was typeset by Typst — the same engine, and the same code path, that CourseC uses to render its own booklets and question papers.*